SSH Key in GitHub

What is SSH?

SSH (Secure Shell) is a cryptographic network protocol used to securely connect one computer to another over an unsecured network.

In the context of **GitHub**, SSH is used to securely authenticate your local machine with your GitHub account without requiring you to enter your username and password every time you push or pull code.

What is an SSH Key?

An SSH key is a pair of cryptographic keys used for secure authentication. It consists of:

1. Private Key

- Stored securely on your local machine
- Must never be shared
- Used to prove your identity

2. Public Key

- Uploaded to your GitHub account
- Can be shared safely
- Used by GitHub to verify your identity

How SSH Authentication Works

When you run a command like:

```
git push
```

The following happens:

1. GitHub sends an encrypted challenge.
2. Your computer uses your **private key** to sign the challenge.
3. GitHub verifies the response using your **public key**.
4. If verification succeeds, access is granted.

This process allows secure authentication without transmitting passwords.

Why Use SSH Instead of HTTPS?

Advantages of SSH:

- More secure authentication method
- No need to repeatedly enter credentials
- Convenient for frequent Git operations
- Ideal for development workflows and CI/CD environments

HTTPS Authentication:

- Requires a username and password or a Personal Access Token
- Less convenient for daily development tasks

Common Types of SSH Keys

The most commonly used types are:

- **RSA**
- **ED25519** (Recommended: faster and more secure)

ED25519 is currently considered best practice for most developers.

How Can You Add SSH Key?

1. Open Git Bash.
2. ssh-keygen -t ed25519 -C mohammedhamdi262003@gmail.com

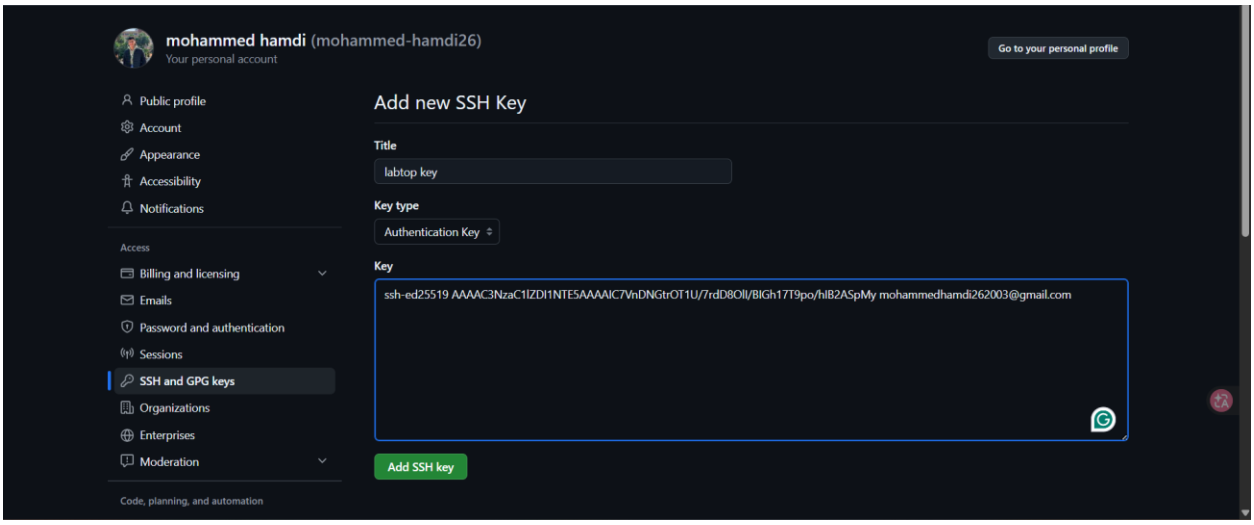
```
$ ssh-keygen -t ed25519 -C "mohammedhamdi262003@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/3MO EL4AMS/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/3MO EL4AMS/.ssh/id_ed25519
Your public key has been saved in /c/Users/3MO EL4AMS/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:4SisCEcg+anf4ikWITta0yvA8tffj34x40f6UlNlRjs mohammedhamdi262003@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|o.             ..|
|o.              =|
| ...    .      E.|
|.oo.   o .      o|
|+oo.o . S      . |
|B=o...      + +  |
|+*oo o     . B . |
|o.=.+ .   . .= .  |
|.oo+   ...+o.+.   |
+----[SHA256]-----+
```

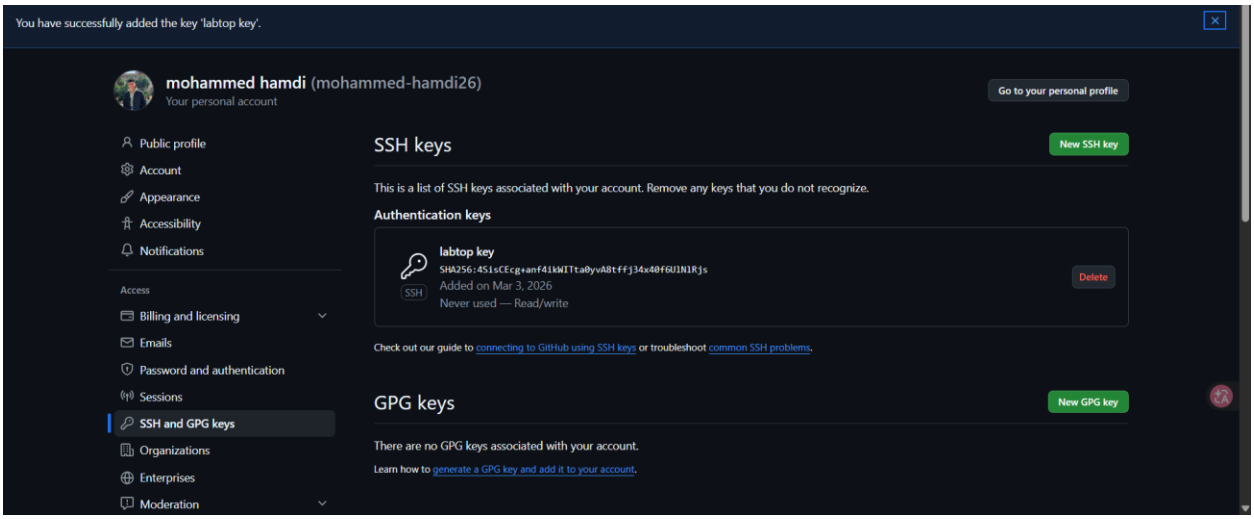
3. Copy the Public Key

```
3MO EL4AMS@mohammed-hamdi MINGW64 ~/Desktop
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIC7VnDNGtrOT1U/7rdD8OlI/BIGh17T9po/hIB2ASpMy mohammedhamdi262003@gmail.com
```

4. **Add the SSH Key to GitHub**

    1. Sign in to your GitHub account.
    2. Go to **Settings**.
    3. Click **SSH and GPG keys**.
    4. Click **New SSH key**.
    5. Add a title (e.g., "My Laptop").
    6. Paste your copied public key.
    7. Click **Add SSH key**.





5. Test the Connection

```
$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi mohammed-hamdi26! You've successfully authenticated, but GitHub does not provide shell access.
```